

Question ID: 7812801f

Question

The fynbos shrubland is a diverse habitat found only in South Africa. It is adjacent to the Afro-temperate forest, with almost no transition space between the two distinct habitats. Plants in the fynbos have uniquely thin and long root systems that spread out over large distances to absorb nutrients from the soil. Ecologists transplanted tree seedlings from the forest into plots in the fynbos. Seedlings in plots isolated from the roots of fynbos plants exhibited a growth rate five times greater than that of the seedlings in plots in close proximity to the roots of fynbos plants.

Based on the text, what role do fynbos roots most likely have in maintaining the border between the fynbos shrubland and the Afro-temperate forest habitats?

Answer

- A. Fynbos roots damage the root systems of forest plants, leaving those plants unable to acquire sufficient nutrients.
- B. Fynbos roots extend close enough to the forest plants' roots that they constitute a physical barrier that forest plants' roots cannot pass.
- C. The root systems of fynbos plants allow the plants to take in so many soil nutrients that forest plants are prevented from flourishing in the fynbos.
- D. The root systems of fynbos plants enhance the soil immediately surrounding the plants, allowing them to thrive in an otherwise harsh habitat.